WORKED SOLUTION

# STEP III 2021, Question 1

Pure •

**STEP III 2021, Question 1**

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

## The key idea

Both parts follow the same recipe: differentiate the parametric equations, take the negative reciprocal of $\mathrm{d}y/\mathrm{d}x$ to get the gradient of the normal, and then tidy the equation. The tidying is where all the fun happens. Multiply through to clear the fractions, and the trigonometric terms collapse into a short, symmetric line equation. Once the line is in the form $x\sin\phi + y\cos\phi = \ldots$ or $x\cos\phi + y\sin\phi = \ldots$, its geometry (intercepts, distance from the origin) can be read off at sight.

## Solution

### Part (i)

Differentiating with respect to $t$,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 12\cos^2 t\,\sin t, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 12\cos t - 12\sin^2 t\cos t = 12\cos t\,(1-\sin^2 t) = 12\cos^3 t.$$

So, for $0 < t < \frac{1}{2}\pi$ (where $\sin t$ and $\cos t$ are nonzero),

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{12\cos^3 t}{12\cos^2 t\sin t} = \frac{\cos t}{\sin t}.$$

At $t = \phi$ the tangent has gradient $\cot\phi$, so the normal has gradient $-\tan\phi = -\dfrac{\sin\phi}{\cos\phi}$.

The normal is therefore

$$y - 12\sin\phi + 4\sin^3\phi = -\frac{\sin\phi}{\cos\phi}(x + 4\cos^3\phi).$$

Multiply by $\cos\phi$ and collect the $x$ and $y$ terms on the left:

$$\begin{aligned} x\sin\phi + y\cos\phi &= 12\sin\phi\cos\phi - 4\sin^3\phi\cos\phi - 4\cos^3\phi\sin\phi \\ &= 12\sin\phi\cos\phi - 4\sin\phi\cos\phi\,(\sin^2\phi + \cos^2\phi) \\ &= 8\sin\phi\cos\phi. \end{aligned}$$

So the normal is

$$\boxed{x\sin\phi + y\cos\phi = 8\sin\phi\cos\phi} \qquad \text{or equivalently} \qquad \frac{x}{\cos\phi} + \frac{y}{\sin\phi} = 8.$$

The second form shows that the normal meets the axes at $(8\cos\phi, 0)$ and $(0, 8\sin\phi)$.

**Tangency.** Let $P = (8\cos^3\phi,\ 8\sin^3\phi)$. We need three things.

*$P$ is on the curve $x^{2/3} + y^{2/3} = 4$.* Since $0 < \phi < \frac{1}{2}\pi$, both coordinates are positive, so the real cube roots are $2\cos\phi$ and $2\sin\phi$. Hence $x^{2/3} + y^{2/3} = 4\cos^2\phi + 4\sin^2\phi = 4$.

*$P$ is on the normal.* Substituting,

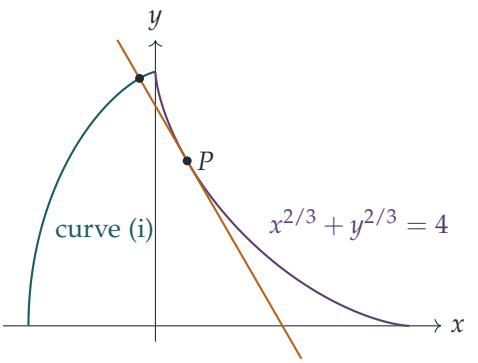
$$8\cos^3\phi\sin\phi + 8\sin^3\phi\cos\phi = 8\sin\phi\cos\phi\,(\cos^2\phi + \sin^2\phi) = 8\sin\phi\cos\phi.$$

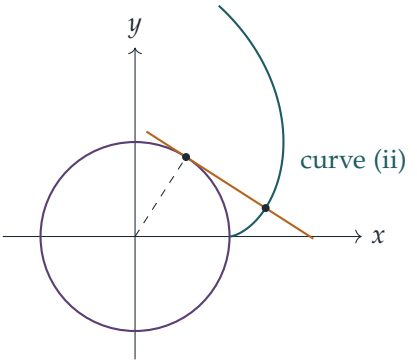
*The gradients agree at $P$.* Differentiating $x^{2/3} + y^{2/3} = 4$ implicitly (valid for $x, y > 0$),

$$\tfrac{2}{3}x^{-1/3} + \tfrac{2}{3}y^{-1/3}\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{y^{1/3}}{x^{1/3}} = -\frac{2\sin\phi}{2\cos\phi} = -\tan\phi \ \text{ at } P,$$

which is exactly the gradient of the normal.

A line through a point of a curve with the same gradient as the curve there is the tangent at that point. So the normal is a tangent to $x^{2/3} + y^{2/3} = 4$ at $P$, as required.




In each picture the normal (amber) at a point of the curve (teal) touches the second curve (plum); both are drawn for one particular value of $\phi$.

#### Part (ii)

Differentiating with respect to $t$ (product rule on the $t\sin t$ and $t\cos t$ terms),

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t + \sin t + t\cos t = t\cos t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \cos t - \cos t + t\sin t = t\sin t.$$

For $0 < t < \frac{1}{2}\pi$ we have $t\cos t \neq 0$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{t\sin t}{t\cos t} = \tan t$. At $t = \phi$ the normal has gradient $-\cot\phi = -\dfrac{\cos\phi}{\sin\phi}$, and its equation is

$$y - \sin\phi + \phi\cos\phi = -\frac{\cos\phi}{\sin\phi}(x - \cos\phi - \phi\sin\phi).$$

Multiplying by $\sin\phi$,

$$y\sin\phi - \sin^2\phi + \phi\sin\phi\cos\phi = -x\cos\phi + \cos^2\phi + \phi\sin\phi\cos\phi.$$

The $\phi\sin\phi\cos\phi$ terms cancel, and $\sin^2\phi + \cos^2\phi = 1$, leaving

$$\boxed{x\cos\phi + y\sin\phi = 1}.$$

**Ask yourself** How do you find the distance from the origin to a line without finding the foot of the perpendicular?

The distance from $(x_0, y_0)$ to the line $ax + by = c$ is $\dfrac{|ax_0 + by_0 - c|}{\sqrt{a^2 + b^2}}$. Here $(x_0, y_0) = (0,0)$, $a = \cos\phi$, $b = \sin\phi$ and $c = 1$, so the distance is

$$\frac{|-1|}{\sqrt{\cos^2\phi + \sin^2\phi}} = 1.$$

You can also see it directly: $(\cos\phi, \sin\phi)$ is a unit vector perpendicular to the line, and the point $(\cos\phi, \sin\phi)$ itself satisfies the equation. So the foot of the perpendicular from $O$ is $(\cos\phi, \sin\phi)$, at distance 1.

**The curve.** Every normal is at distance 1 from the origin, whatever $\phi$ is. A line at distance 1 from $O$ meets the circle of radius 1 about $O$ only at the foot of the perpendicular, and there it is perpendicular to the radius. That makes it a tangent to the circle. So every normal is a tangent to

$$\boxed{x^2 + y^2 = 1},$$

touching it at $(\cos\phi, \sin\phi)$.

## Checks

- In (i) take $\phi = \frac{1}{3}\pi$. The curve point is $(-\frac{1}{2},\ 6\sqrt{3} - \frac{3}{2}\sqrt{3}) = (-0.5,\ 7.794)$ and the normal $2x + \frac{2}{\sqrt{3}}y = 8$ meets the axes at $(4,0)$ and $(0, 4\sqrt{3}) \approx (0, 6.928)$. The point $P = (1,\ 3\sqrt{3}) \approx (1,\ 5.196)$ satisfies $2 + 6 = 8$, so it lies on the normal, and $1^{2/3} + (3\sqrt{3})^{2/3} = 1 + 3 = 4$.
- In (ii), at $\phi \to 0$ the curve starts at $(1,0)$ on the unit circle, and the normal there tends to the vertical line $x = 1$, which is indeed a tangent to $x^2 + y^2 = 1$.
- A computer check over a thousand random values of $\phi$ confirmed all the normal equations, the tangency in (i) and the unit distance in (ii) to within rounding.

## Common slips

- Getting $\mathrm{d}y/\mathrm{d}t$ wrong in (i): $12\cos t - 12\sin^2 t\cos t$ simplifies to $12\cos^3 t$, and the factorisation is what makes the gradient so simple.
- Only checking that $P$ lies on the normal in (i). A line through a point of a curve is not automatically a tangent there; the gradients must match too.
- Answering "a circle" in (ii) without giving its equation, centre and radius.

## Beyond STEP

The envelope of all the normals to a curve is called its *evolute*. It is the locus of the centres of curvature. So part (i) shows that the astroid $x^{2/3} + y^{2/3} = 4$ is the evolute of the first curve. In (ii) the curve is the *involute* of the unit circle (the path traced by the end of a taut string unwound from a reel), and its evolute is the circle. Involutes of circles give the profile of most gear teeth. These ideas belong to the differential geometry of curves. Needham, *Visual Complex Analysis*, is a beautifully geometric companion to this kind of reasoning about plane curves.

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